Ahmed Elnour

Interactive Developer

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Profile

Interactive Developer building with **Unity**, AR and hardware, creating **interactive installations** driven by depth camera body tracking, DMX and Art-Net lighting and Arduino controls, alongside mobile AR apps and games. 5+ years of Unity and game development experience, with a Bachelor's degree in Electrical and Electronics Engineering and a strong focus on performance, maintainability and production efficiency.

Skills

Interactive Installations: Body tracking gesture input, touchscreen kiosk applications, physical controls and lighting driven from Unity

Hardware: Arduino, motors, DMX and Art-Net lighting, Astera LED tubes, Orbbec Femto Bolt depth cameras, OpenCV

Unity3D: Proficient in Unity3D development, including game mechanics, UI design, AR development and more

Unity Tools: Scriptable Objects, Addressables, JSON, shaders, Cinemachine, particles, WebGL (personal projects)

Performance and Stability: Unity Profiler, Firebase Crashlytics, memory optimization, debugging and unit testing

Programming: C#, GDScript, Java, Python, Bash and Node.js for game logic, scripting, automation and SSH servers

Multiplayer: Realtime multiplayer with PUN2 (Photon Unity Networking), including matchmaking, rooms and game state sync

Godot and Git: Godot Engine (2D/3D, GDScript, editor tools); Git and GitHub in solo and team projects

Languages: Arabic (native), English (proficient)

Experience

Rayqube Sep 2026 – Oct 2026
Interactive Developer (Contract) *Dubai, UAE*

Unity Development

- **Body Tracking Gesture Input:** Built a Unity 6 gesture recognition system on **Orbbec Femto Bolt** depth cameras, turning skeleton data into game actions (punches, weaves and ducks) for an interactive installation.
- **Accuracy Testing and Tuning:** Measured detection accuracy against hand labelled video recordings using a record and replay pipeline, then tuned thresholds and filters, cutting false detections by over 70%. Used the Unity Profiler to analyze performance.
- **Touchscreen Kiosks:** Built touchscreen kiosk applications in Unity.
- **Reusable Packages and Tooling:** Packaged the camera and gesture systems as reusable Unity packages with editor setup checks, a diagnostics overlay, keyboard simulation for testing without a camera, and integration guides for developers and AI coding agents.

Mechatronics and Operations

- **Lighting Control:** Controlled **Astera** LED tubes and other fixtures over **DMX** and **Art-Net** for interactive installations.
- **LED Color Calibration:** Analyzed video content against the color gamut of RGB LED tubes and designed a calibrated color conversion that brought cue colors to a barely visible difference.
- **Physical Controls:** Connected **Arduino** based physical buttons to Unity to control the app's behavior.
- **Hardware Configuration:** Used Scriptable Objects to configure hardware parameters and LED sequences.
- **Remote On-site Support:** Supported on-site installations remotely using AnyDesk and GitHub.

Sabil Systems Inc. Dec 2024 – Present
Software Engineer *Edmonton, Canada*

- **Architected Scalable Game Systems:** Designed robust, extensible architectures in Unity (C#) and Godot (GDScript), converting abstract concepts into technical specifications for long term product lifecycles.
- **Pipeline Innovation:** Developed custom Unity based tools and reusable game templates, significantly reducing designer iteration time and streamlining production pipelines.
- **Profiling and Crash Reporting:** Used the Unity Profiler to analyze performance and Firebase Crashlytics for crash reporting.

- **Achievements, Settings and Store Content:** Worked on the achievements system, managed settings with Scriptable Objects and JSON, and delivered remote store content with JSON.
- **Established Technical Standards:** Authored technical roadmaps and enforced rigorous code review guidelines, ensuring high codebase stability and team wide technical excellence.
- **UI and Visual Polish:** Engineered responsive, high performance mobile UIs and utilized advanced particle systems and shaders to enhance player feedback and game feel.

Capital Digital Mar 2024 – Dec 2024
Game Developer *Riyadh, Saudi Arabia*

- **Dynamic AI Architecture:** Engineered an AI system for player vs computer gameplay, utilizing state machines to create adaptive and challenging opponent behavior.
- **Core Loop Engineering:** Developed the foundational gameplay mechanics and main game loop, ensuring high performance synchronization between game state and user input.
- **UX Driven Development:** Designed and implemented a responsive UI framework and an interactive tutorial system for intuitive onboarding.

Interactive Future Aug 2023 – Feb 2024
Game Developer *Riyadh, Saudi Arabia*

- **Native Android Integration:** Developed and bridged native Android modules with the game engine to leverage platform specific features and optimize mobile performance.
- **Extensible Tooling:** Created custom libraries, plugins, and editor tools that accelerated the internal development cycle and standardized feature implementation.
- **Rapid Prototyping:** Iterated on prototypes to validate game mechanics, translating successful proofs of concept into production ready C# code.

CTRL A Studios Apr 2022 – Jan 2023
Game Developer *Khartoum, Sudan*

- **Systems Implementation:** Delivered core gameplay features and 2D/3D visual systems, collaborating closely with artists to integrate complex animations, VFX and Cinemachine cameras.
- **Quality Assurance and Support:** Conducted debugging and unit testing to eliminate critical glitches, while providing technical support and documentation for team members.

The Code Effect Dec 2021 – Dec 2022
AR Developer *Khartoum, Sudan*

- **AR Systems Engineering:** Designed and maintained Augmented Reality applications using **AR Foundation**, focusing on stable spatial tracking, plane and image recognition, and interactive animations.
- **Fullstack AR Integration:** Connected the AR frontend with **Flutter based** mobile apps, backend servers and databases for realtime data persistence, with a local Node.js backend for testing.

Projects

AR Cheese Hunt GitHub
Mobile AR Game *Unity, AR Foundation, Android*

- **AR Game Loop:** Built an AR game that turns the player's room into the playfield: the player anchors a character to a detected plane with a raycast, then drives it with an on-screen joystick to collect cheese that spawns across tracked surfaces in a 60-second scoring loop.
- **Playtest Iteration:** Replaced a world-space arrow with a screen-space pointer after playtesters kept losing the character off-camera.

Sky Hopper Google Play
Shipped Mobile Game *Unity, C#*

- **Shipped Game:** Designed and shipped a 3D puzzle platformer with swipe based movement, tiles that crumble moments after contact, 12 hand crafted levels and beat the clock scoring of up to 3 stars.

Education

University of Khartoum 2018
Bachelor of Electrical and Electronics Engineering *Khartoum, Sudan*